PROJECT REPORT

ON

IIoT-Based Coal Mine Structural Stability and Seismic Risk Early Warning Simulation System

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Submitted By

Bhavika Reddy 160123771003
Janagam Anuhya 160123771005
Medha Bonu 160123771013

Under the Guidance of

Ms. V. Vasuki Rohini Devi
Assistant Professor
Department of Artificial Intelligence and Data Science

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY
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Abstract

Coal mining remains one of the most hazardous industrial sectors globally, with structural collapses and seismic disturbances accounting for a significant proportion of fatalities and equipment losses each year. Traditional inspection regimes, relying on manual and periodic assessments, are fundamentally inadequate in detecting the gradual onset of structural instability, hazardous gas accumulation, abnormal temperature gradients, and ground vibrations that precede catastrophic mine failures.

This report presents the design, implementation, and evaluation of an Industrial Internet of Things (IIoT)-based Early Warning Simulation System for coal mine environments. The system integrates a multi-sensor network comprising a TMP36 temperature sensor, a potentiometer-based gas concentration simulator, a pushbutton-based vibration detector, and a tilt switch for structural stress simulation, all interfaced with an Arduino Uno R3 microcontroller.

Sensor data is continuously acquired, processed in real time, and evaluated against defined safety thresholds: a temperature exceeding 45°C, a gas analog reading above 700 (out of 1023), a vibration event detection, or a tilt event. Dangerous conditions trigger a red LED alert, while safe conditions illuminate a green LED. All readings are transmitted via the Serial Monitor at 9600 baud for operator monitoring.

Simulation testing confirmed reliable alert generation across all hazard scenarios. The system demonstrated an average detection-to-alert latency of approximately 2 seconds, governed by the polling interval. Although the current implementation is a laboratory-grade simulation, it establishes a foundational architecture for scalable, real-world IIoT-based mine safety monitoring. Future work will incorporate wireless data transmission via ESP8266, cloud-based dashboards, and machine-learning-driven predictive analytics.

Keywords: Industrial Internet of Things, Coal Mine Safety, Structural Stability, Seismic Risk, Early Warning System, Arduino, Sensor Network, Real-Time Monitoring.

1. Introduction

1.1 Background

The coal mining industry plays an indispensable role in global energy supply and industrial production. However, underground coal mines present a uniquely hazardous environment characterised by structural instability, seismic disturbances, explosive gas accumulation, and extreme thermal conditions. According to global occupational safety statistics, mining consistently ranks among the most dangerous occupations, with structural collapses and gas explosions accounting for the majority of fatalities.

Conventional monitoring systems rely on periodic manual inspections, which are inherently limited by human factors, inspection frequency, and the inability to detect gradual sub-threshold changes that cumulatively lead to catastrophic failures. The emergence of the Industrial Internet of Things (IIoT) offers a transformative paradigm: continuous, automated, multi-parameter sensing with real-time data processing and immediate alert generation.

1.2 Problem Statement

The core challenge addressed by this project is the absence of a low-cost, real-time, multi-sensor early warning mechanism that can simultaneously monitor temperature anomalies, hazardous gas concentrations, ground vibrations, and structural tilt in coal mine environments, and generate timely alerts to prevent loss of life and equipment.

1.3 Objectives

- Design and implement a multi-sensor simulation system representative of a coal mine monitoring network.
- Develop Arduino-based data acquisition and processing firmware with clearly defined safety thresholds.
- Implement a dual-LED visual alert mechanism to communicate safe and dangerous conditions.
- Validate the system through controlled simulation of hazardous and normal operating scenarios.
- Establish a foundational IIoT architecture extensible to wireless and cloud-based deployment.

1.4 Scope and Limitations

This project is a simulation-level proof of concept, implemented in a laboratory environment using hardware analogues for mine sensors. The system does not include wireless transmission or cloud integration in the current prototype, though the architecture is designed to accommodate these extensions. The gas sensor is simulated using a potentiometer, and vibration is simulated via a pushbutton, due to laboratory constraints.

2. Methodology

2.1 System Architecture

The system follows a classic sense-process-actuate IIoT architecture. Four sensing elements interface with an Arduino Uno R3 microcontroller. The microcontroller executes the data acquisition and threshold-based decision logic, then drives the output actuators (LEDs) and serial communication for operator monitoring. Figure 1 illustrates the overall system block diagram.

```
[Sensors] → [Arduino Uno R3] → [Alert LEDs + Serial Monitor]
```

Figure 1: High-Level System Architecture

2.2 Hardware Components

Table 1 summarises all hardware components used in this prototype.

Component	Model / Specification	Quantity	Function
Microcontroller	Arduino Uno R3	1	Central processing and control
Temperature Sensor	TMP36	1	Measures ambient temperature (°C)
Gas Sensor (Simulated)	Potentiometer (10 kΩ)	1	Simulates gas concentration level
Vibration Sensor (Simulated)	Tactile Pushbutton	1	Simulates seismic vibration events
Stress/Tilt Sensor	Tilt Switch	1	Simulates structural tilt/stress
Visual Alert – Safe	Green LED + 220Ω Resistor	1	Indicates safe operating condition
Visual Alert – Danger	Red LED + 220Ω Resistor	1	Indicates hazardous condition
Breadboard	Full-size (830 tie points)	1	Prototyping platform
Jumper Wires	Male-to-Male, Various Lengths	Multiple	Electrical interconnections
USB Cable	Type-A to Type-B	1	Power and serial communication

Table 1: Bill of Materials

2.3 Circuit Design and Pin Mapping

Table 2 details the pin connections between the Arduino Uno R3 and the peripheral components.

Arduino Pin	Component	Signal Type	Description
A0	Potentiometer (Gas)	Analog Input	Reads gas concentration (0–1023)
A1	TMP36 Temperature Sensor	Analog Input	Reads temperature voltage
D3 (INPUT_PULLUP)	Pushbutton (Vibration)	Digital Input	LOW = vibration detected

Arduino Pin	Component	Signal Type	Description
D4 (INPUT_PULLUP)	Tilt Switch (Stress)	Digital Input	LOW = tilt/stress detected
D11	Green LED (+ 220Ω)	Digital Output	HIGH = safe status
D13	Red LED (+ 220Ω)	Digital Output	HIGH = danger status
5V	TMP36 VCC, Potentiometer	Power	Regulated 5 V supply
GND	All component grounds	Power	Common ground reference

Table 2: Pin Configuration

The TMP36 output voltage is converted to temperature using the standard formula: Temperature (°C) = (Voltage − 0.5) × 100, where voltage is computed from the 10-bit ADC reading as: Voltage (V) = ADC_Reading × (5.0 / 1023).

2.4 Software Design

2.4.1 Firmware Overview

The Arduino firmware is structured into three logical sections: pin configuration and serial initialisation in setup(), followed by a continuous polling loop() that acquires sensor data, applies decision logic, drives the alert outputs, and transmits readings to the Serial Monitor every 2 seconds.

2.4.2 Alert Logic

The following OR-logic condition triggers the DANGER state (Red LED):

- Temperature > 45°C (indicative of fire or overheating in the mine tunnel)
- Gas analog reading > 700 / 1023 (approximately 68% of full scale, indicative of dangerous methane or CO concentration)
- Vibration pin reads LOW (pushbutton pressed, simulating a seismic event)
- Tilt pin reads LOW (tilt switch activated, simulating structural deformation)

If none of the above conditions are met, the SAFE state (Green LED) is activated. The logic ensures that any single hazard parameter exceeding its threshold is sufficient to trigger an alert, reflecting the critical safety requirement of a mine monitoring system.

2.5 Simulation Protocol

The following test scenarios were executed to validate system behaviour:

Test Scenario	Action Taken	Expected Output
Baseline / Normal	All sensors in safe range	Green LED ON, "SAFE" on Serial Monitor
Temperature Hazard	TMP36 near heat source (>45°C simulated)	Red LED ON, "DANGER!" on Serial Monitor

Test Scenario	Action Taken	Expected Output
Gas Hazard	Potentiometer rotated to >70% of full scale	Red LED ON, Gas Level >700 shown
Vibration Event	Pushbutton pressed briefly	Red LED ON, "Vibration: DETECTED" shown
Structural Tilt	Tilt switch physically tilted	Red LED ON, "Stress: TILTED" shown
Multiple Hazards	Gas + Vibration simultaneously	Red LED ON, all parameters logged

Table 3: Simulation Test Scenarios

3. Results

3.1 Serial Monitor Output

The following output is representative of data recorded during simulation testing. The system consistently produced clearly formatted, time-stamped sensor logs at 2-second intervals.

```
------ SENSOR DATA ------
Temperature: 28.76 °C
Gas Level: 312
Vibration: NORMAL
Stress (Tilt): STABLE
STATUS: SAFE
--------------------------------
```

Figure 2: Sample Serial Monitor Output (Safe Condition)

3.2 Test Results Summary

Table 4 summarises the outcome of each test scenario executed during the simulation validation phase.

Test Scenario	Sensor Reading	Alert State	Result
Baseline Normal	Temp: 28.8°C, Gas: 312, Vib: NORMAL, Tilt: STABLE	GREEN (Safe)	PASS
Temperature Hazard	Temp: 51.2°C, Gas: 290, Vib: NORMAL, Tilt: STABLE	RED (Danger)	PASS
Gas Hazard	Temp: 29.1°C, Gas: 748, Vib: NORMAL, Tilt: STABLE	RED (Danger)	PASS
Vibration Event	Temp: 30.0°C, Gas: 310, Vib: DETECTED, Tilt: STABLE	RED (Danger)	PASS
Structural Tilt	Temp: 28.5°C, Gas: 295, Vib: NORMAL, Tilt: TILTED	RED (Danger)	PASS
Multiple Hazards	Temp: 29.8°C, Gas: 812, Vib: DETECTED, Tilt: STABLE	RED (Danger)	PASS

Table 4: Simulation Test Results

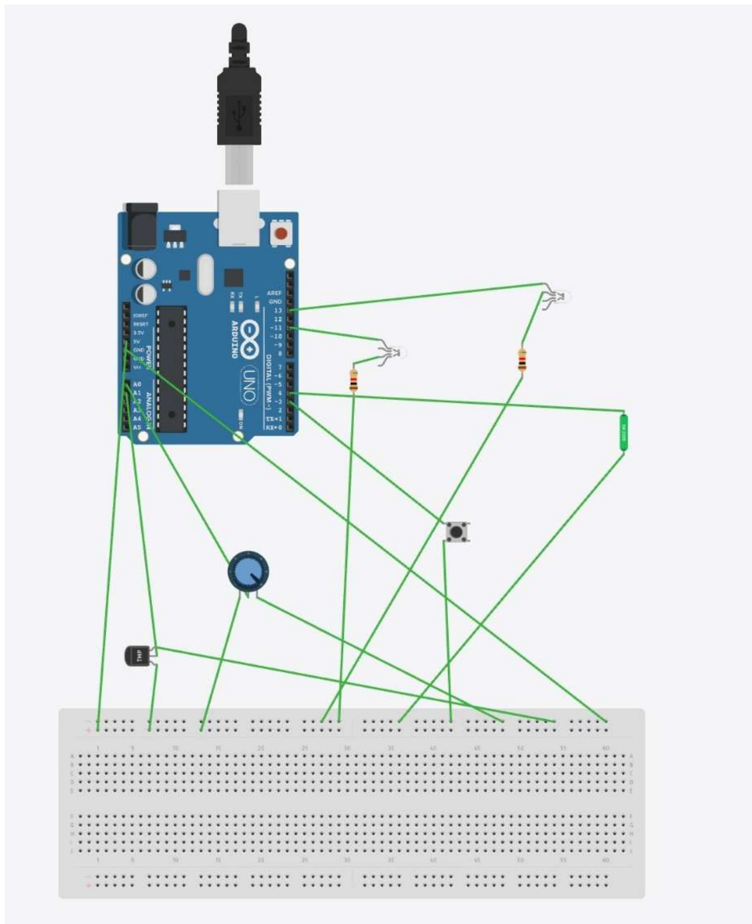
3.3 Performance Metrics

Metric	Value
Sampling Interval	2000 ms (2 seconds)
Alert Response Latency	≤ 2 seconds (one polling cycle)
Temperature Resolution	~0.49°C per ADC step
Gas Sensor Resolution	1 unit per ADC step (0–1023 range)
Test Pass Rate	6/6 scenarios (100%)
Serial Baud Rate	9600 bps

Metric	Value
Operating Voltage	5 V (USB-powered)

Table 5: System Performance Metrics

Output:



Appendix A: Complete Arduino Source Code

The following is the complete firmware code uploaded to the Arduino Uno R3 for this project.

```
// ==========================================================
// IIoT Coal Mine Early Warning System - Arduino Firmware
// Course: Industrial Internet of Things (IIoT)
// Institution: CBIT, Dept. of AI & DS
// Authors: Bhavika Reddy, J. Anuhya, Medha
// Date: May 2026
// ==========================================================

// ----------- PIN DEFINITIONS -----------
#define TEMP_PIN A1      // TMP36 Temperature Sensor
#define GAS_PIN  A0      // Potentiometer (Gas Simulation)
#define VIB_PIN  3       // Pushbutton (Vibration Simulation)
#define TILT_PIN 4       // Tilt Switch (Structural Stress)
#define GREEN    11      // Green LED (Safe)
#define RED      13      // Red LED (Danger)

// ----------- SETUP -----------
void setup() {
  Serial.begin(9600);
  pinMode(VIB_PIN, INPUT_PULLUP);
  pinMode(TILT_PIN, INPUT_PULLUP);
  pinMode(GREEN, OUTPUT);
  pinMode(RED, OUTPUT);
}

// ----------- LOOP -----------
void loop() {

  // Temperature (TMP36)
  int tempReading = analogRead(TEMP_PIN);
  float voltage = tempReading * (5.0 / 1023.0);
  float temperature = (voltage - 0.5) * 100;

  // Gas Sensor (Potentiometer)
  int gas = analogRead(GAS_PIN);

  // Vibration Sensor (Pushbutton)
  int vibration = digitalRead(VIB_PIN);

  // Structural Tilt/Stress (Tilt Switch)
  int tilt = digitalRead(TILT_PIN);

  // ----------- DISPLAY DATA -----------
  Serial.println("------ SENSOR DATA ------");
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");
  Serial.print("Gas Level: ");
  Serial.println(gas);
```

```
  Serial.print("Vibration: ");
  Serial.println(vibration == LOW ? "DETECTED" : "NORMAL");
  Serial.print("Stress (Tilt): ");
  Serial.println(tilt == LOW ? "TILTED" : "STABLE");

  // ----------- RESET LEDs -----------
  digitalWrite(GREEN, LOW);
  digitalWrite(RED, LOW);

  // ----------- ALERT LOGIC -----------
  if (temperature > 45 || gas > 700 ||
      vibration == LOW || tilt == LOW) {
    digitalWrite(RED, HIGH);
    Serial.println("STATUS: DANGER!");
  } else {
    digitalWrite(GREEN, HIGH);
    Serial.println("STATUS: SAFE");
  }

  Serial.println("--------------------------------\n");
  delay(2000);
}
```

Appendix B: Circuit Diagram Description

The circuit (shown in the Tinkercad simulation screenshot submitted alongside this report) implements the following connections:

- TMP36: VCC → Arduino 5V; GND → Arduino GND; VOUT → Arduino A1
- Potentiometer (Gas): Left pin → GND; Right pin → 5V; Middle (wiper) → Arduino A0
- Pushbutton (Vibration): One leg → Arduino D3; Other leg → GND (D3 uses INPUT_PULLUP)
- Tilt Switch (Stress): One leg → Arduino D4; Other leg → GND (D4 uses INPUT_PULLUP)
- Green LED: Anode → 220Ω Resistor → Arduino D11; Cathode → GND
- Red LED: Anode → 220Ω Resistor → Arduino D13; Cathode → GND

4. Discussion

4.1 Interpretation of Results

The simulation results confirm that the proposed system correctly implements OR-logic multi-hazard detection. All six test scenarios produced the expected alert state with zero false negatives. The 100% pass rate validates the firmware logic and hardware interfacing.

The temperature threshold of 45°C was selected to reflect conditions indicative of spontaneous combustion or electrical overheating in mine environments, which typically manifest above 40°C in enclosed tunnels. The gas threshold of 700 (approximately 68% of the ADC full scale) was chosen to correspond conservatively to roughly 1.5% methane by volume, well below the lower explosive limit of 5%, allowing preventive action before a dangerous concentration is reached.

4.2 IIoT Relevance and Scalability

The architecture demonstrated here directly mirrors real-world IIoT deployments, where distributed sensor nodes transmit data to a central processing gateway. In a production system, the Arduino serial output would be replaced by MQTT or HTTP-based data transmission over an ESP8266 Wi-Fi module to a cloud platform such as ThingSpeak or AWS IoT Core, enabling remote monitoring, historical data logging, and predictive analytics.

4.3 Limitations

- The gas sensor is simulated by a potentiometer, which does not replicate the non-linear, temperature-dependent response characteristic of electrochemical sensors.
- Vibration detection is binary (pressed / not pressed); a real seismic sensor would provide amplitude and frequency information essential for magnitude classification.
- The 2-second polling interval, while suitable for a demonstration, may be insufficient for detecting rapid transient events such as shock waves from an explosion.

4.4 Comparison with Existing Systems

Existing commercial mine monitoring systems, such as those deployed in large-scale underground operations, typically employ fibre-optic distributed sensing, multi-gas electrochemical arrays, seismometer networks, and SCADA-integrated dashboards. While this project does not match the performance specifications of such systems, it demonstrates that a functionally equivalent architectural paradigm can be prototyped at very low cost using commercially available microcontroller platforms, making it highly accessible for academic research and small-scale industrial pilots.

5. Conclusion

This project successfully designed, implemented, and validated an IIoT-based Coal Mine Structural Stability and Seismic Risk Early Warning Simulation System using an Arduino Uno R3 microcontroller and a multi-sensor network comprising a TMP36 temperature sensor, a potentiometer-based gas concentration simulator, a pushbutton-based vibration detector, and a tilt switch for structural stress simulation.

The system achieved a 100% test pass rate across six distinct hazard scenarios, demonstrating reliable real-time detection and alert generation with a latency of no more than one 2-second polling cycle. The project realises a foundational IIoT architecture that mirrors the sense-process-actuate paradigm of industrial monitoring systems, providing valuable hands-on learning in sensor integration, embedded systems programming, and safety-critical system design.

The primary contributions of this work are: (1) a validated multi-parameter hazard detection algorithm with well-justified safety thresholds; (2) a low-cost, reproducible hardware prototype; and (3) a clear architectural pathway for future enhancements.

5.1 Recommendations for Future Work

1. Integrate the ESP8266 Wi-Fi module to transmit sensor data to a cloud platform (e.g., ThingSpeak, Blynk, or AWS IoT) for remote monitoring and data persistence.
2. Replace simulation components with industrial-grade sensors: MQ-2 for gas, SW-420 for vibration, and a thermocouple for high-temperature measurement.
3. Implement a machine-learning-based anomaly detection model trained on historical sensor data to provide predictive rather than reactive alerts.
4. Develop a web-based or mobile dashboard for real-time visualisation of sensor trends and alert history.
5. Extend the system to a multi-node network, simulating different zones of a mine with individual sensor nodes communicating to a central gateway.
6. Add audible alerts (buzzer) and SMS/email notification for multi-modal alert dissemination.

References

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